

[CMP370] **Internet of Things**

Unit 4: Developing IoT System

Interfacing to connect with external devices using development boards



Role of Microcontrollers (Raspberry Pi)

- Acts as the central processing unit in IoT systems, handling data acquisition, processing, and communication.
- Raspberry Pi's GPIO pins enable interfacing with sensors/actuators, while its networking capabilities (Wi-Fi, Ethernet) connect to the cloud.

Introduction to Python for IoT:

- **Basics of Python:** Syntax, variables, loops, functions.
- **Libraries:**
 - **RPi.GPIO:** Controls GPIO pins (e.g., turning on an LED).
 - **smbus:** Communicates with I2C devices (e.g., sensors).
 - **spidev:** Interfaces with SPI devices (e.g., displays).
- Python's simplicity makes it ideal for IoT prototyping.

Setting Up the Raspberry Pi for IoT

- Install Raspberry Pi OS (e.g., via Raspberry Pi Imager).
- Configure network settings (Wi-Fi, SSH for remote access).
- Install Python and required libraries (pip install RPi.GPIO).

Connecting Smart Objects

IoT devices and sensors must be connected to the network for their data to be utilized.

“Communications Criteria,” describes the characteristics and attributes you should consider when selecting and dealing with connecting smart objects.

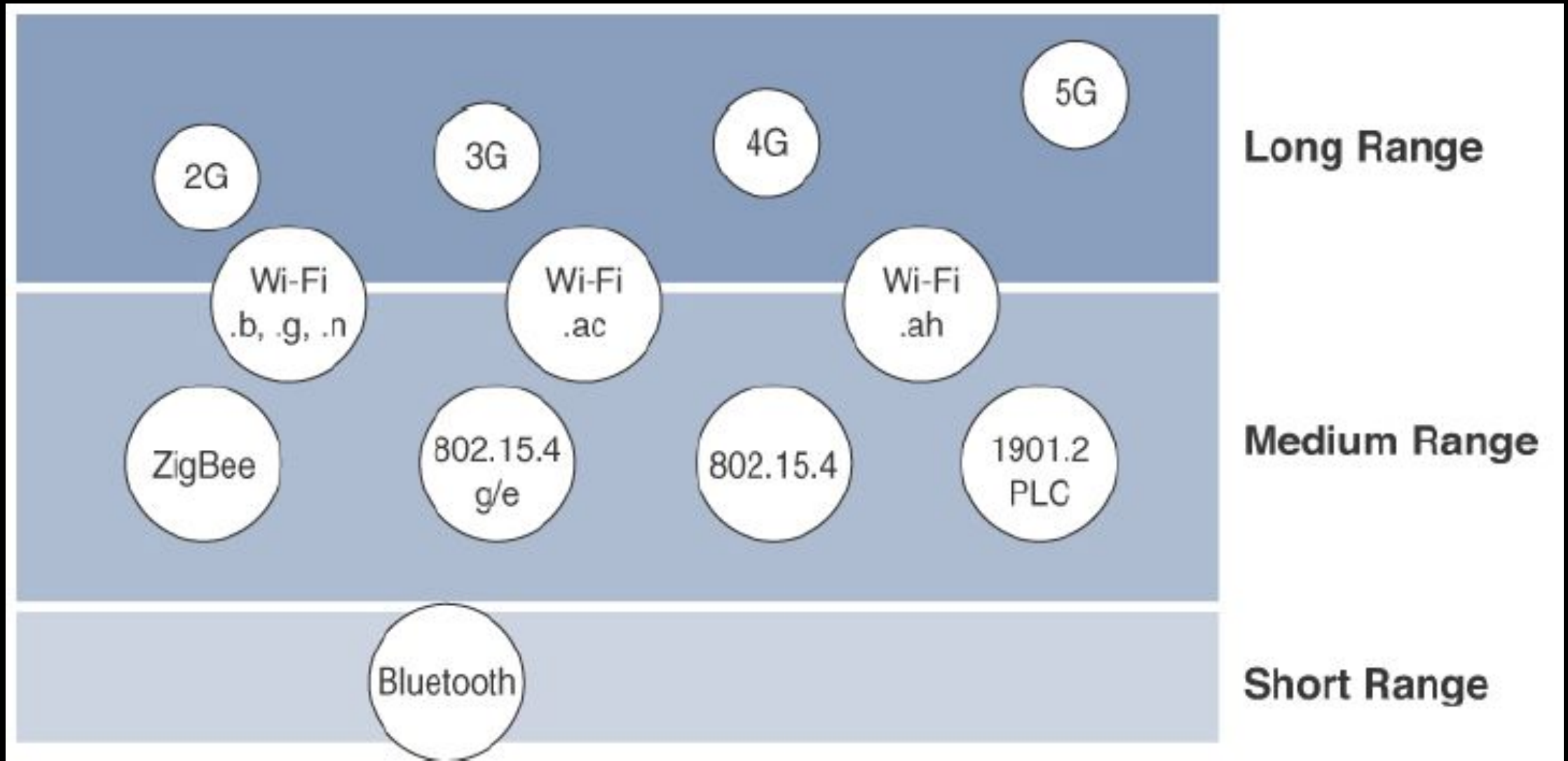
- a) Range
- b) Frequency Bands
- c) Power Consumption
- d) Topology
- e) Constrained Devices.
- f) Constrained-Node Networks.

IoT Access Technologies

Some of the technologies that are considered when connecting smart objects

- a) Standardization and alliances:** The standards bodies that maintain the protocols for a technology
- b) Physical layer:** The wired or wireless methods and relevant frequencies
- c) MAC layer:** Considerations at the Media Access Control (MAC) layer, which bridges the physical layer with data link control
- d) Topology:** The topologies supported by the technology
- e) Security:** Security aspects of the technology
- f) Competitive technologies**

Wireless Access Landscape



Range

How far does the signal need to be propagated?

Should indoor versus outdoor deployments be differentiated?

□ **Short range:** The classical wired example is a serial cable. Wireless short-range technologies are often considered as an alternative to a serial cable, supporting tens of meters of maximum distance between two devices. Examples: IEEE 802.15.1 Bluetooth and IEEE 802.15.7 Visible Light Communications (VLC).

Medium Range

□ **Medium range:** This range is the main category of IoT access technologies.

- In the range of tens to hundreds of meters, many specifications and implementations are available.
- The maximum distance is generally less than 1 mile between two devices.
- Examples: IEEE 802.11 Wi-Fi, IEEE 802.15.4, and 802.15.4g WPAN

Long Range

- **Long range:** Distances greater than 1 mile between two devices require long-range technologies.
- Wireless examples are cellular (2G, 3G, 4G) and some applications of outdoor IEEE 802.11 Wi-Fi and Low-Power Wide-Area (LPWA) technologies.
- LPWA communications have the ability to communicate over a large area without consuming much power.
- These technologies are therefore ideal for battery powered IoT sensors.

Frequency Bands

- Radio spectrum is regulated by countries and/or organizations, such as the International Telecommunication Union (ITU) and Nepal Telecommunication Authority (NTA).
- These groups define the regulations and transmission requirements for various frequency bands. For example, portions of the spectrum are allocated to types of telecommunications such as radio, television, military, and so on.
- Focusing on IoT access technologies, the frequency bands leveraged by wireless communications are split between licensed and unlicensed bands.

Frequency Bands

- Licensed spectrum is generally applicable to IoT long-range access technologies and allocated to communications infrastructures deployed by services providers, public services (for example, first responders, military), broadcasters, and utilities.
- An important consideration for IoT access infrastructures that wish to utilize licensed spectrum is that users must subscribe to services when connecting their IoT devices. This adds more complexity to a deployment involving large numbers of sensors and other IoT devices, but in exchange for the subscription fee, the network operator can guarantee the exclusivity of the frequency usage over the target area and can therefore sell a better guarantee of service. Improvements have been made in handling the complexity that is inherent

Unlicensed spectrum

- The ITU has also defined unlicensed spectrum for the industrial, scientific, and medical (ISM) portions of the radio bands. These frequencies are used in many communications technologies for short-range devices (SRDs). Unlicensed means that no guarantees or protections are offered in the ISM bands for device communications. For IoT access, these are the most well known ISM bands:
 - ❖ 2.4 GHz band as used by IEEE 802.11b/g/n Wi-Fi
 - ❖ IEEE 802.15.1 Bluetooth
 - ❖ IEEE 802.15.4 WPAN

Unlicensed spectrum

- Some communications within the ISM bands operate in the sub-GHz range. Sub-GHz bands are used by protocols such as IEEE 802.15.4, 802.15.4g, and 802.11ah, and LPWA technologies such as LoRa and Sigfox.
- The disadvantage of sub-GHz frequency bands is their lower rate of data delivery compared to higher frequencies. However, most IoT sensors do not need to send data at high rates. Therefore, the lower transmission speeds of sub-GHz technologies are usually not a concern for IoT sensor deployments.

Power Consumption

- While the definition of IoT device is very broad, there is a clear delineation between powered nodes and battery-powered nodes.
- A powered node has a direct connection to a power source, and communications are usually not limited by power consumption criteria. However, ease of deployment of powered nodes is limited by the availability of a power source, which makes mobility more complex.
- Battery-powered nodes bring much more flexibility to IoT devices.
- IoT wireless access technologies must address the needs of low power consumption and connectivity for battery-powered nodes. This has led to the evolution of a new wireless environment known as Low-Power Wide-Area (LPWA).

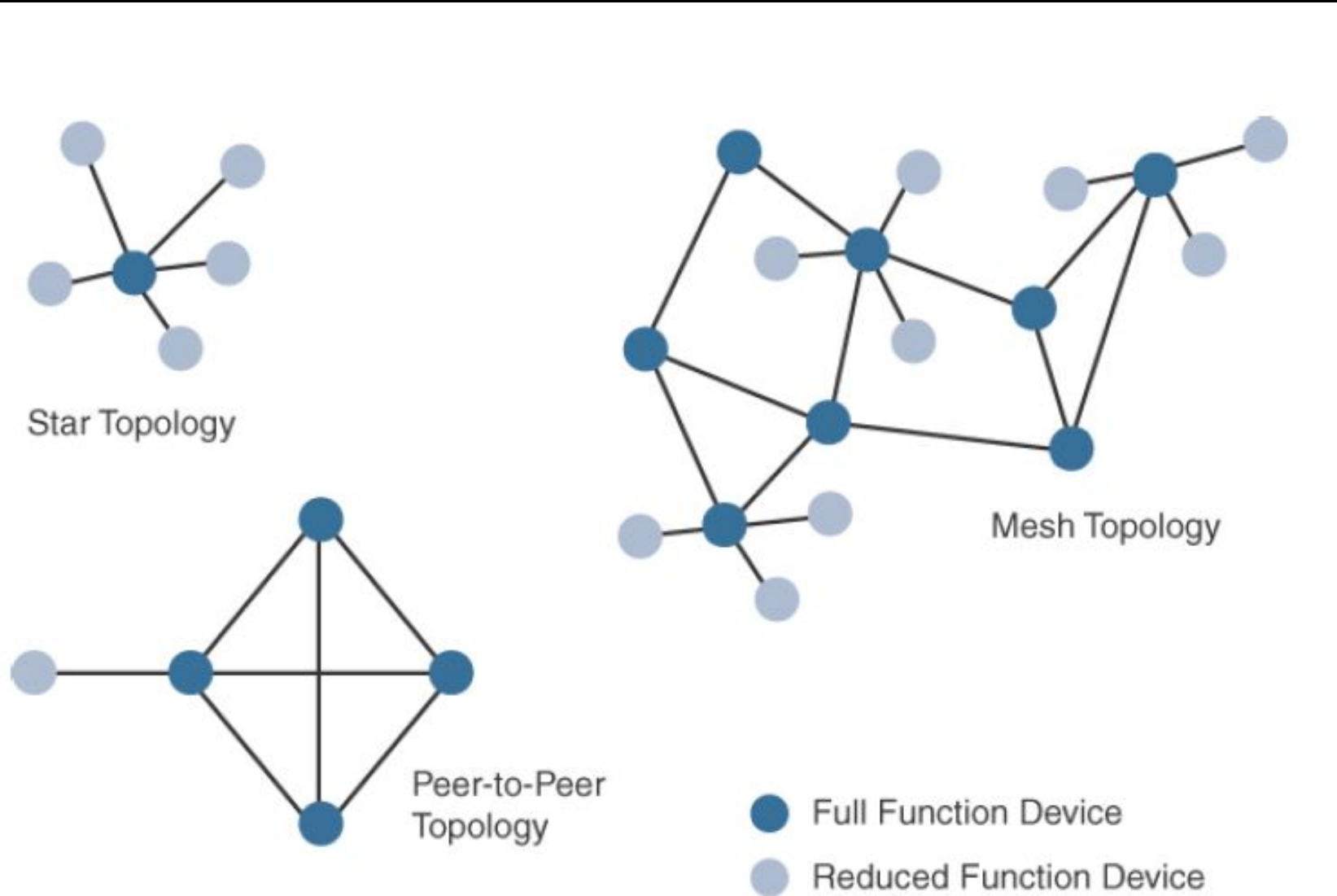
Topology

- Among the access technologies available for connecting IoT devices, three main topology schemes are dominant: star, mesh, and peer-to-peer.
- For long range and short-range technologies, a star topology is prevalent, as seen with cellular, LPWA, and Bluetooth networks. Star topologies utilize a single central base station or controller to allow communications with endpoints.
- For medium-range technologies, a star, peer-to-peer, or mesh topology is common,

Topology

- Peer-to-peer topologies allow any device to communicate with any other device as long as they are in range of each other. Obviously, peer-to-peer topologies rely on multiple full-function devices.
- Peer-to-peer topologies enable more complex formations, such as a mesh networking topology.

Topology



Sensor Interfacing with Python

- Connect sensors to GPIO pins (e.g., DHT11 for temperature/humidity).
- Write Python code to read sensor data (e.g., using `adafruit_dht` library).
- Example: Reading temperature every 10 seconds and printing to the console.

Actuator Interfacing with Python

- Connect actuators to GPIO pins (e.g., a relay to control a fan).
- Write Python code to control actuators (e.g., RPi.GPIO to toggle a relay).
- Example: Turning on a fan if the temperature exceeds 30°C.

Constrained Devices

- The Internet Engineering Task Force (IETF) acknowledges in RFC 7228 that different categories of IoT devices are deployed.
- While categorizing the class of IoT nodes is a perilous exercise, with computing, memory, storage, power, and networking continuously evolving and improving, RFC 7228 gives some definitions of constrained nodes.
- According to RFC 7228, constrained nodes can be broken down into the three classes

Class	Definition
Class 0	This class of nodes is severely constrained, with less than 10 KB of memory and less than 100 KB of Flash processing and storage capability. These nodes are typically battery powered. They do not have the resources required to directly implement an IP stack and associated security mechanisms. An example of a Class 0 node is a push button that sends 1 byte of information when changing its status. This class is particularly well suited to leveraging new unlicensed LPWA wireless technology.
Class 1	While greater than Class 0, the processing and code space characteristics (approximately 10 KB RAM and approximately 100 KB Flash) of Class 1 are still lower than expected for a complete IP stack implementation. They cannot easily communicate with nodes employing a full IP stack. However, these nodes can implement an optimized stack specifically designed for constrained nodes, such as Constrained Application Protocol (CoAP). This allows Class 1 nodes to engage in meaningful conversations with the network without the help of a gateway, and provides support for the necessary security functions. Environmental sensors are an example of Class 1 nodes.
Class 2	Class 2 nodes are characterized by running full implementations of an IP stack on embedded devices. They contain more than 50 KB of memory and 250 KB of Flash, so they can be fully integrated in IP networks. A smart power meter is an example of a Class 2 node.

Constrained-Node Networks

- While several of the IoT access technologies, such as Wi-Fi and cellular, are applicable to laptops, smart phones, and some IoT devices, some IoT access technologies are more suited to specifically connect constrained nodes.
- Typical examples are IEEE 802.15.4 and 802.15.4g RF, IEEE 1901.2a PLC, LPWA, and IEEE 802.11ah access technologies.
- Constrained-node networks are often referred to as low-power and lossy networks (LLNs).
- Low-power in the context of LLNs refers to the fact that nodes must cope with the requirements from powered and battery-powered constrained nodes.
- Lossy networks indicates that network performance may suffer from interference and variability due to harsh radio environments.

Data Analytics

- Data analytics is a widely used term, which can refer to specific methods or to general processes.
- The term data refers to the facts, figures, statistics collected and processed with the aim to extract meaningful information from it. In, general, data can be anything, for example, weather data, the revenue of a company over the years, time series, images ,text ,and speech, etc.
- The intention behind processing and extracting meaningful information is to understand the behavior and to benefit the related entity.
- Analytics refers to the systematic, mathematical, and logical analysis of data.

Data Logging

- Data logging is the process of collecting and storing data over a period of time in different systems or environments. It involves tracking a variety of events.
- It is collecting data about a specific, measurable topic or topics, regardless of the method used.
- In modern industries, data is most often collected and stored using a data logger.
- A data logger is an electronic sensor device with a microprocessor and a memory. Typically, the data is then manually transferred to a computer by retrieving the device and plugging it into the computer's USB port for data downloads, though it can also be transferred to a cloud system.
- Cloud databanks are the most efficient option for storing the collected data in most use cases.

How does data logging work

- The data logging process consists of four main steps:
 - A. A sensor gathers and records the data from one or more sources.
 - B. A microprocessor then performs basic measurement and logic tasks, such as adding, subtracting, transferring and comparing numbers.
 - C. Data stored in the memory unit of the data logger is then transferred to a computer or other electronic device for analysis.
 - D. Once analyzed, the data is visualized through a knowledge graph or chart.

Types of data loggers

A. Standalone data loggers

Standalone data loggers, or standalone sensors, are small, portable devices typically equipped with a USB port. These devices can either have an internal or external sensor which allows the device to track data from an on-site or remote location, respectively.

B. Wireless data loggers

Wireless loggers, or wireless sensors, are a type of standalone data logger that accesses data via wireless technology (such as a mobile app or Bluetooth) and transfers it via cloud technology. This eliminates the need for manually retrieving and compiling data from various systems

Types of data loggers

C. Computer-based data loggers

As the name implies, computer-based data loggers, or computer-based sensors, are data loggers that are tethered to a computer. A computer-based logger supports real-time visibility into sensor data, while software applications on the computer enable real-time analysis. The main drawback of a computer-based logger is that it is limited by the system the sensor can run on.

D. Web-based data loggers

Web-based data loggers, or web-based sensors, are the most advanced type of data logger. This system is connected to the internet, typically through a wireless network; though, in some cases, an ethernet connection may still be used. Collected data is transferred and stored on a remote server and accessed on demand.

Data Visualization

- Internet of Things (IoT) has become one of the most emerging and powerful technologies that is used to improve the quality of life.
- IoT connects together a great number of heterogeneous devices in order to dynamically acquire various types of data from the real-world environment.
- IoT data is used to mine useful information that may be used, by context-aware applications, in order to improve people's daily life.
- As data is typically featured with contextual information (time, location, status, etc), IoT turns into a valuable and voluminous source of contextual data with variety (several sources), velocity (real-time collection), veracity (uncertainty of data) and value.

Data Visualization

- With the advancements in the deep architectures, it is now possible to develop systems that are capable of learning from high-dimensional data(millions of parameters).
- However, in order to get more insights into the data/learned parameters, good visualization methods are required.
- Such visualizations help in a deeper understanding of how these architectures are working, giving an edge to improvement in architectural development.

Data Visualization

- The cooperation of Big Data and IoT has initiated the development of smart services for many complex infrastructures.
- As IoT develops rapidly, Big Data technologies play a critical role, as visual analytics tools, producing valuable knowledge in real-time, within the IoT infrastructures, aiming in supporting critical decision making.
- Visual analytics is an analysis technique that can assist the exploration of vast amounts of data by utilizing data mining, statistics, and visualization.
- Interactive visualization tools combine automated analysis and human interaction allowing user control during the data analysis process, aiming in producing valuable insight for decision making.

Data Visualization

- Visualizing data plays a major role in data analytics since it manifests the presentation of findings and its patterns concurrently with the original data.
- Data visualization helps to interpret the results by correlating the findings to the goals. It also exposes hidden patterns, trends, and correlations, that otherwise would be undetected, in an impactful and perceptible manner.

Types of Charts

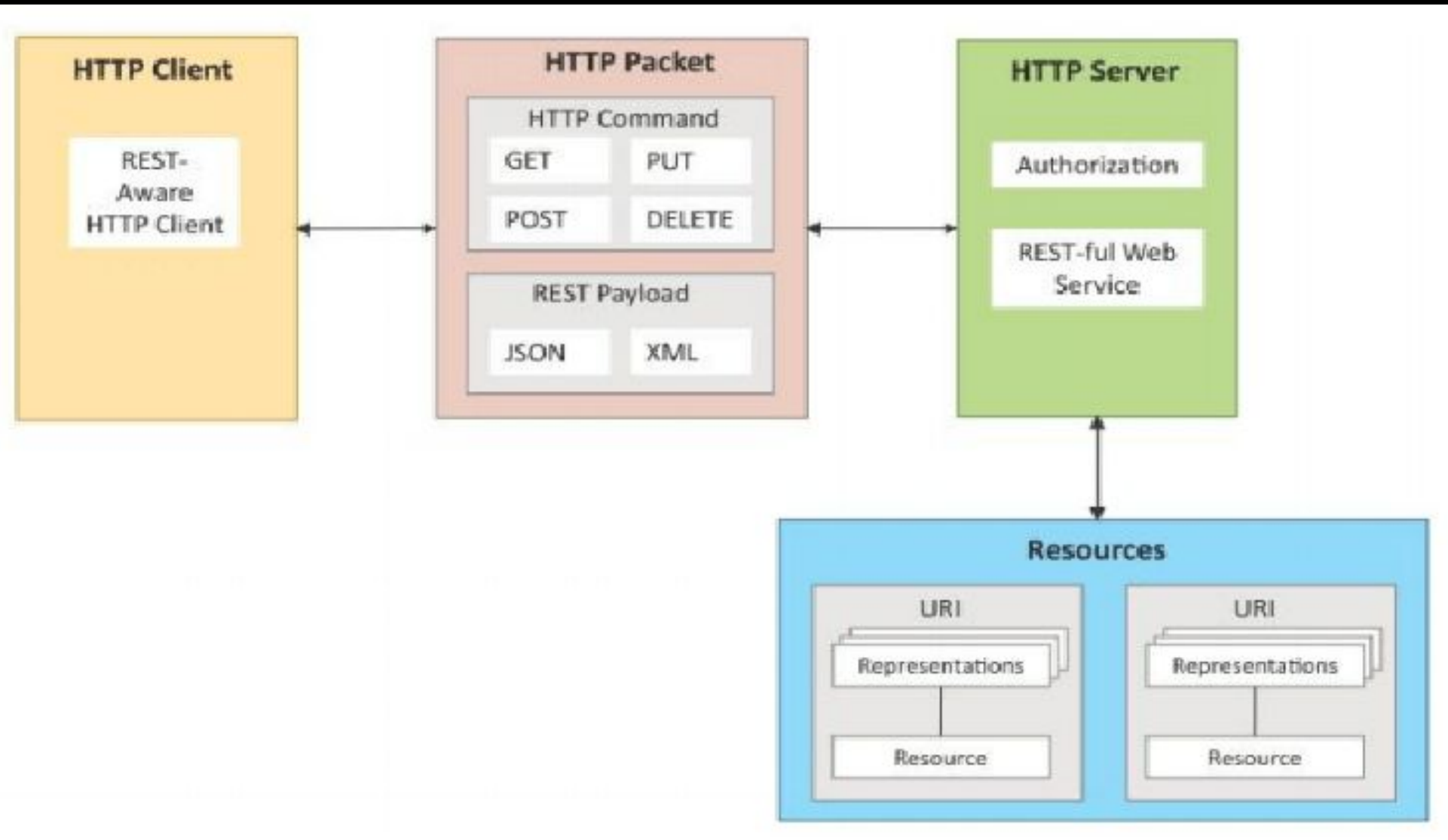
- In the lifeline of data visualization, data scientists operate various visualization tools that implement many conventional visualization techniques involving numerous kinds of plots and charts.
- Lines-plots are continuous curves that may be used to depict functional dependencies between output variables.
- Radial-charts show the different extents of the relative values that each variable can take.
- Maps are useful for identifying the relative difference or behavior in different parts of data or if the data has geographic information.
- Circular heatmaps are used to compactly depict the change of a variable over a month
- Diagrams and Graphs are also extensions of charts used in special cases.

IoT Communication APIs

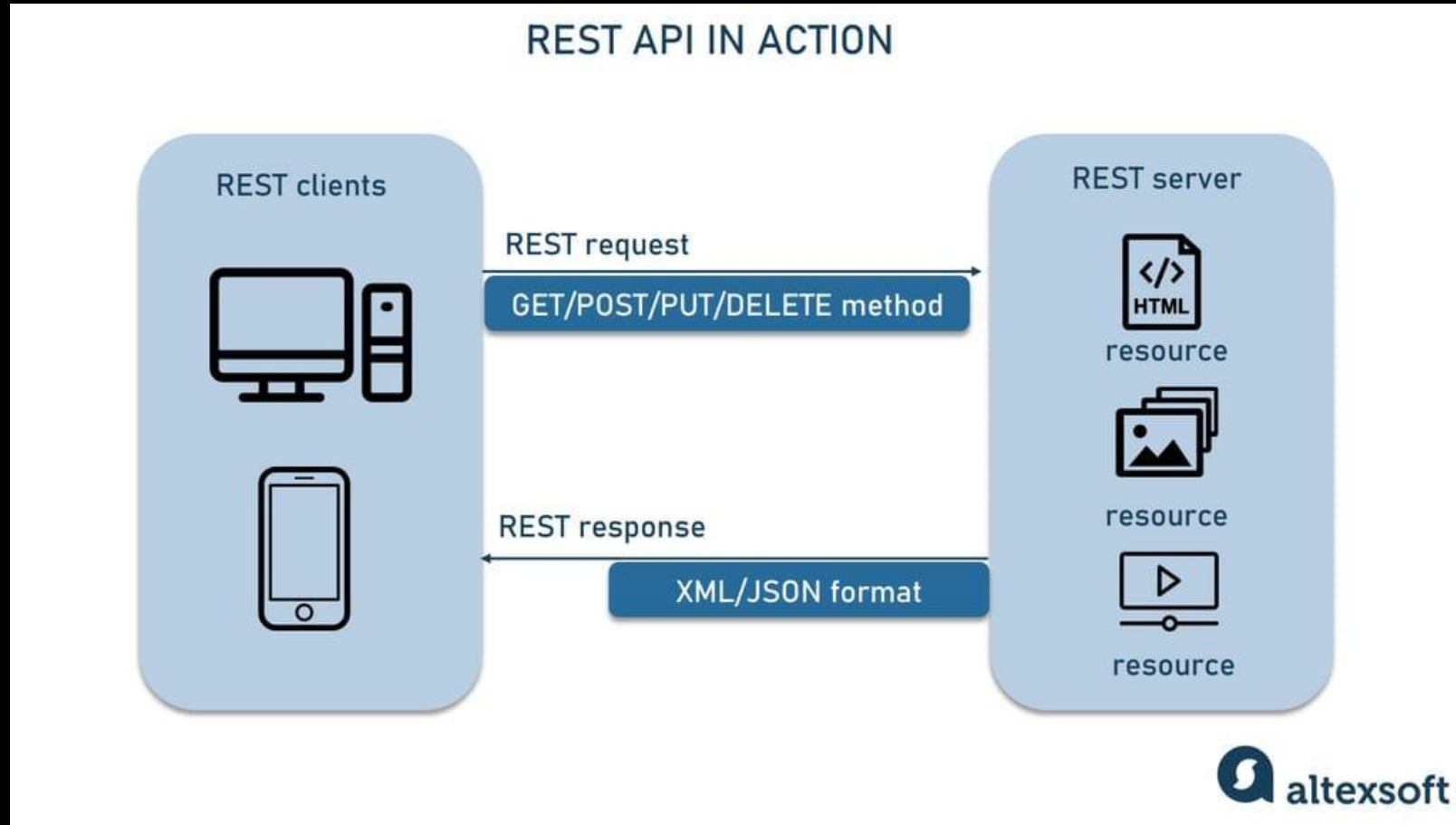
REST-based Communication APIs

- Representational State Transfer (REST) is a set of architectural principles by which you can design web services and web APIs that focus on a system's resources and how resource states are addressed and transferred. REST APIs follow the request-response communication model.
- The REST architectural constraints apply to the components, connectors, and data elements, within a distributed hypermedia system.

Communication with REST APIs



Request-response model used by REST



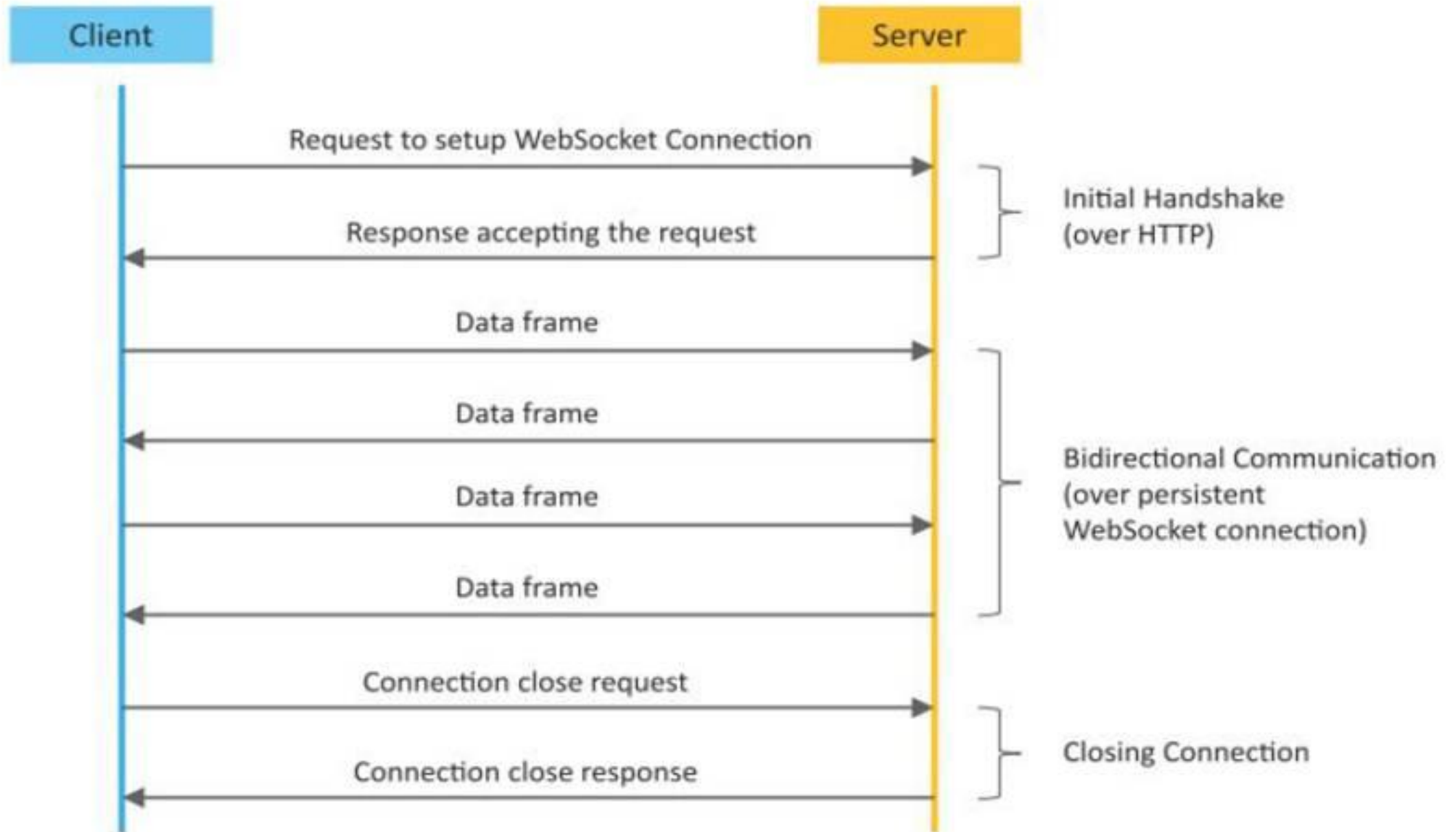
REST-based Communication APIs

- A RESTful web service is a "web API" implemented using HTTP and REST principles.
- RESTful web service is a collection of resources which are represented by URIs. RESTful web API has a base URI (e.g. `http://example.com/api/tasks/`).
- The clients send requests to these URIs using the methods defined by the HTTP protocol (e.g., GET, PUT, POST, or DELETE).
- A RESTful web service can support various Internet media types (JSON being the most popular media type for RESTful web services).

WebSocket-based Communication APIs

- WebSocket APIs allow bi-directional, full duplex communication between clients and servers.
- WebSocket APIs follow the exclusive pair communication model
- Unlike request-response APIs such as REST, the WebSocket APIs allow full duplex communication and do not require a new connection to be setup for each message to be sent. WebSocket communication begins with a connection setup request sent by the client to the server.
- This request (called a WebSocket handshake) is sent over HTTP and the server interprets it as an upgrade request. If the server supports WebSocket protocol, the server responds to the WebSocket handshake response. After the connection is setup, the client and server can send data/messages to each other in full-duplex mode.
- WebSocket APIs reduce the network traffic and latency as there is no overhead for connection setup and termination requests for each message.

WebSocket Protocol



IoT Communication (MQTT)

- Use MQTT to send sensor data to a broker (e.g., Mosquitto).
- Write Python code to publish data (e.g., using paho-mqtt library) and subscribe to commands.
- Example: Publishing temperature readings to a topic and subscribing to a “fan on/off” command.

Message Queue Telemetry Transport (MQTT)

- (MQTT) is a commonly used protocol in IoT. IBM developed this for the purpose of machine-to-machine communication. This is kind of lightweight security protocol; it works on top of the TCP layer.
- The request and response time is minimized; network bandwidth consumption is very less. MQTT maintains message information, and it acts as broker.
- As a broker, it creates topic based on user data, when user sends message from client machine to MQTT broker; the component automatically replies based on the topic. MQTT works based on Internet bandwidth and battery level.

MQTT

- Message Queuing Telemetry Transport (MQTT) is basically a network protocol to transfer data between devices. MQTT is built on the principle of publish-subscribe pattern.
- MQTT can be used for communications with embedded devices like IoT sensors.
- Indeed, the MQTT protocol is efficient for devices powered by battery because it does not use a large bandwidth and accepts high latency.
- Originally, MQTT has been used for communications between devices on Earth and satellites.

Case Study:

Home Automation, Weather Monitoring System

- **Home Automation:** Control lights and appliances using sensors (e.g., motion sensor to turn on lights) and MQTT for remote control.
- **Weather Monitoring System:** Use temperature, humidity, and pressure sensors to monitor environmental conditions, visualize data on a dashboard, and send alerts if thresholds are crossed.

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